

AI Training Program for Working Professionals

A practical, example-driven course in Prompt Engineering, AI Office Productivity, AI-Assisted Software Development, and GitHub Copilot.

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Day 1	Prompt Engineering · AI for Office Productivity
Day 2	AI-Assisted Software Development · GitHub Copilot
Format	4 sessions · approx. 90 minutes each · 6 hours total
Level	Working professionals with basic AI-tool exposure

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How to Use This Book

This book is the complete study material for a 2-day, 4-session AI training program. It is written in simple, plain English on purpose, with short sentences and real company examples, so it is easy to follow for any audience, including non-native English speakers.

Who this course is for

This course is designed for working professionals who already have some basic experience with AI tools such as ChatGPT, but who have not yet used AI in a structured, work-focused way. No programming background is required for Sessions 1 and 2. Sessions 3 and 4 introduce coding concepts gently, with plenty of pictures and no assumption of deep technical knowledge.

The 6-hour schedule

Day & Time	Session	Focus
Day 1 08:45 – 10:30	Session 1	Prompt Engineering
Day 1 10:45 – 12:15	Session 2	AI for Office Productivity
Day 2 08:45 – 10:30	Session 3	AI for Software Development & AI-Assisted Coding
Day 2 10:45 – 12:15	Session 4	GitHub Copilot

Sessions 1 and 3 run 105 minutes; Sessions 2 and 4 run 90 minutes. Each chapter in this book is written with a flexible core (fits 90 minutes) plus clearly marked extension material (fills the extra 15 minutes in Sessions 1 and 3). The schedule can be adapted to a single intensive day, a half-day workshop, or spread across more sessions, as needed.

What you will be able to do after this course

- Write clear, structured prompts that get useful answers from any AI chat tool, on the first or second try.
- Use AI inside everyday office tools (Word, Excel, PowerPoint, Outlook/Gmail, Teams/Zoom) to save hours every week.
- Understand, at a practical level, how AI assists software developers across the whole development lifecycle.
- Set up and use GitHub Copilot, and explain its main features and pricing plans to your own team.

- Choose the right AI tool for a task from a tested list of free and paid options.

How each chapter is organised

Every session in this book follows the same structure, so you always know where to look:

- **Core concepts** — explained in plain language, with a diagram for every major idea.
- **Real-world examples** — drawn from well-known international brands (Amazon, eBay, Rakuten, Mercari, Sumitomo Corporation, JCB, NTT Docomo, Fujitsu, Sony) with verifiable, public statistics.
- **Classroom exercise** — a short, hands-on activity. Each exercise includes **two versions**: one for a room with laptops and internet, and one that works with pen and paper only, so facilitators can run it either way.
- **Tool directory** — a practical table of free and paid tools for that topic.
- **Key takeaways** — a short summary box to revise before moving on.

FACILITATOR TIP

Encourage questions in any language. If a participant is more comfortable asking a question in their native language, a neighbour can help translate, or the facilitator can restate the idea with an even simpler example. The goal is understanding, not perfect English.

A NOTE ON THE EXAMPLES IN THIS BOOK

Company names, product names and statistics used in this book are drawn from public sources such as company press releases, conference presentations and verified news reports current as of 2026. AI products and pricing change quickly — facilitators should confirm current prices from the official websites before quoting exact numbers to participants.

01

SESSION 1 · DAY 1, 08:45–10:30

Prompt Engineering

How to talk to an AI model so it gives you exactly what you need

BY THE END OF THIS SESSION, YOU WILL BE ABLE TO

- Explain, in simple terms, how a large language model turns your words into an answer.
- Build a prompt using the R-C-T-F framework (Role, Context, Task, Format).
- Apply five core prompting techniques: zero-shot, few-shot, chain-of-thought, role prompting and iterative refinement.
- Rewrite a weak, vague prompt into a strong, specific one.

1 · Prompt Engineering

1.1 Why Prompt Engineering Matters

Every AI chat tool — ChatGPT, Claude, Gemini, Microsoft Copilot — is only as useful as the instruction you give it. Two people can use the exact same AI model and get completely different quality of results, purely because of how they ask. This skill of asking well has a name: **prompt engineering**.

Think of an AI model as a brilliant new employee on their first day. They are extremely well-read and fast, but they do not know your company, your preferences, or what "good" looks like for you, unless you tell them. A vague instruction to a new employee ("make this better") produces a vague result. A clear instruction ("shorten this report to one page, keep the three key numbers, and write it for a non-technical manager") produces a useful result. The AI model works exactly the same way.

QUICK COMPARISON

Weak prompt: "Write about our new product."

Strong prompt: "You are a product marketing writer. Write a 100-word announcement for our new wireless earphones, for a 25–35 year old audience, in an upbeat but professional tone, highlighting battery life and price."

The second prompt takes 15 extra seconds to type and saves 10 minutes of re-writing.

This matters even more for a non-native English speaker. A well-structured prompt does not need perfect grammar — it needs clear structure. In this session, you will learn a simple structure that works every time, in any language, for any task.

Think about it

1. Think of one task you did last week that took longer than it should have (an email, a report, a summary). How might a clear prompt have helped?

1.2 How a Language Model Actually Works

You do not need to be a computer scientist to prompt well, but a simple mental model helps enormously. Here is the one-paragraph version:

A large language model (LLM) is a program trained on an enormous amount of text — books, websites, articles, code. During training, it learns the statistical patterns of language: given the words so far, what word is likely to come next? When you send a prompt, the model reads your entire message as "the words so far" and then predicts the most likely continuation, one small piece at a time, very quickly. It repeats this until it produces a full answer.

Three practical consequences of this design

- **It has no memory beyond what you tell it (in a single chat).** If you do not include the background information the model needs, it will guess — and sometimes guess wrong. This is why **Context** is one of the four building blocks you will learn in Section 1.3.
- **It can "sound" confident even when it is wrong.** Because it is predicting plausible-sounding text, it can generate incorrect facts, numbers, or dates in a fluent, convincing way. This is commonly called a **hallucination**. Always double-check facts, figures and citations before using AI output for anything important.
- **It has a limited "context window."** This is the maximum amount of text (your prompt plus any documents plus its own reply) it can consider at once. For very long documents, you may need to break the task into smaller parts.

IMPORTANT: AI IS A TOOL, NOT AN AUTHORITY

Treat AI output the way you would treat a draft from a junior colleague: useful as a starting point, always reviewed by a human before it is sent to a client, published, or used to make a decision.

1.3 The R-C-T-F Framework

Of all the prompting frameworks in use today, the simplest and most reliable for daily work is **R-C-T-F**: Role, Context, Task, Format. You do not need to use all four labels every time — but thinking through all four before you type will transform your results.

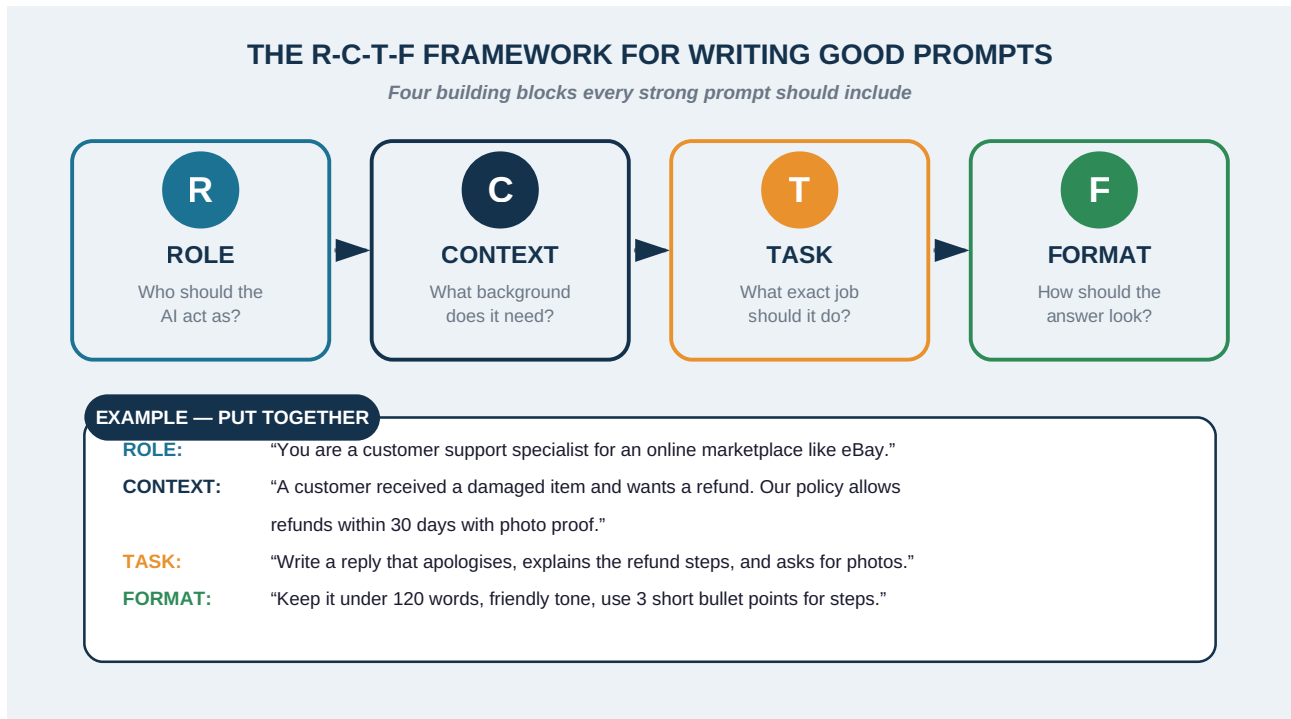


Figure 1.1 — The R-C-T-F framework: four questions to answer before you write a prompt.

Role — who should the AI act as?

Giving the AI a role focuses its "voice" and its knowledge. "You are a senior HR manager" produces a different, more appropriate answer than no role at all, for an HR question.

Context — what background does it need?

This is the most frequently skipped, and most important, ingredient. Include relevant facts: your audience, your constraints, your company's situation, relevant policy, or pasted source text.

Task — what exact job should it do?

Use an action verb and be specific: "summarise," "translate," "compare," "draft a reply," "find errors in," "turn this into a table." Avoid vague verbs like "help with" or "look at."

Format — how should the answer look?

Specify length, structure, tone, and language. "Under 150 words," "3 bullet points," "a formal tone," "in both English and Spanish" all remove guesswork from the model's response.

YOU DO NOT ALWAYS NEED ALL FOUR

For a quick, simple question ("what is the capital of Australia?"), no framework is needed. R-C-T-F earns its value as the task gets more important or more specific — a client email, a report, a presentation, a policy document.

1.4 Core Prompting Techniques

Beyond structure, five techniques consistently improve AI output quality. Use them individually or combine them.

1. Zero-shot prompting

Asking the model to do a task directly, with no examples. Fast, and works well for common, well-understood tasks. Example: "Translate this sentence into French."

2. Few-shot prompting

Giving the model one or two examples of the exact style or format you want before asking it to continue. This is extremely powerful when the "right answer" is hard to describe in words but easy to show.

EXAMPLE: FEW-SHOT FOR CUSTOMER REPLIES

"Here are two examples of our brand's reply style:

Example 1 — Customer: 'My order is late.' Reply: 'We're sorry for the delay — here's your updated tracking link, and a 10% discount on your next order.'

Example 2 — Customer: 'The size is wrong.' Reply: 'No problem at all — here is a free return label, and we'll ship the correct size today.'

Now write a reply in the same style for: 'The product arrived damaged.'"

3. Chain-of-thought prompting

Asking the model to "think step by step" before giving a final answer. This is especially useful for calculations, logical problems, or any task with several steps, because it reduces careless mistakes.

4. Role (persona) prompting

Covered in R-C-T-F above — worth repeating because it is one of the single highest-impact additions to any prompt.

5. Iterative refinement

Nobody writes the perfect prompt on the first attempt. Treat prompting as a short conversation, not a single command.

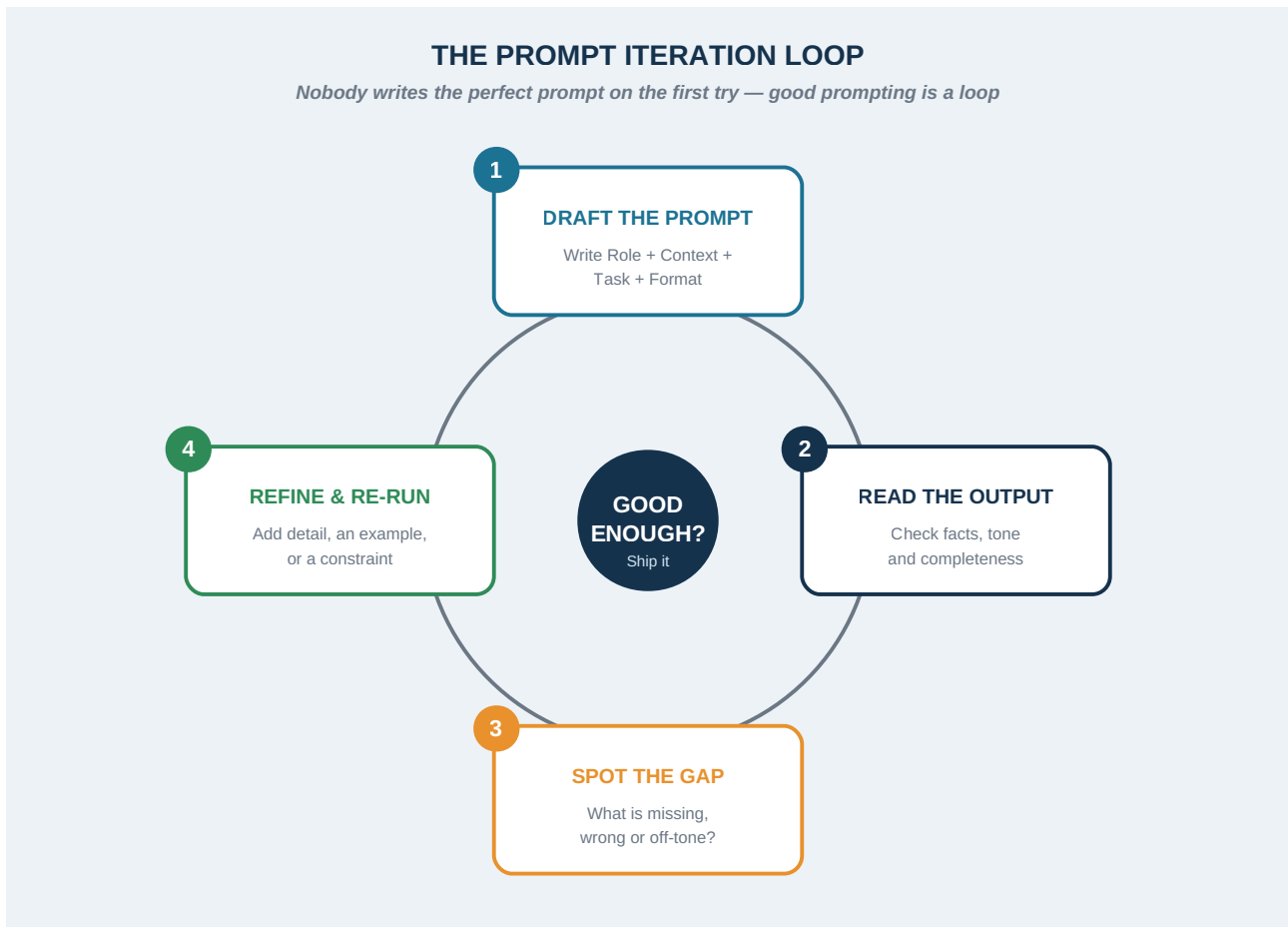


Figure 1.2 — The prompt iteration loop: draft, read, spot the gap, refine, repeat.

FAST REFINEMENT PHRASES

Instead of re-writing your whole prompt, try short follow-ups: "Make it shorter," "Make it more formal," "Add one example," "Explain like I am new to this topic," "Give me three alternative versions."

1.5 Real-World Case Studies

Amazon — Rufus and structured writing culture

Amazon's shopping assistant, **Rufus**, answers customer questions about products directly on the Amazon app by combining product data, reviews and general knowledge — a large-scale example of context-rich prompting happening behind the scenes every time a customer asks a question.

Separately, Amazon is well known internally for its "**working backwards**" writing culture, where employees draft a structured press-release-style document before building a product. This same discipline — being explicit about audience, purpose and outcome before you start — is exactly what the R-C-T-F framework asks you to do with an AI prompt.

eBay — the "Magical Listing" tool

eBay built an AI feature, nicknamed **Magical Listing**, that writes a product title, description, category and suggested price from a seller's photo and a few basic details. In its first phase, roughly 30% of U.S. sellers used it, and more than 95% of sellers who tried the AI-written description chose to keep it, sometimes after light editing. This is a clear, large-scale example of AI reducing a tedious writing task — exactly the kind of task-plus-format prompting you will practise in this session's exercise.

Rakuten — AI for one of the world's largest online marketplaces

Rakuten, a global e-commerce and internet-services group, has invested heavily in generative AI across its businesses, including AI-assisted customer support and product recommendations within its "Rakuten Ichiba" marketplace. Rakuten is a useful reference point for any multilingual organisation: the same prompting discipline used to write an eBay listing in English works identically for writing a marketplace listing in any other language.

WHY THIS MATTERS FOR YOU

Mercari and Rakuten, like eBay, face the same "cold start" listing problem that AI has helped solve. If your organisation sells or lists products online, the same R-C-T-F prompt structure used for a product description works for an internal report, a customer email, or a meeting summary.

EXERCISE 1 **Fix the Weak Prompt****Duration:** 15 minutes • Individual or pairs

The scenario: Your manager asks you to "use AI to write something for our clients about our new service." A colleague tries this prompt:

"Write something about our new service for clients."

Your task: Rewrite this into a strong prompt using R-C-T-F. Decide: who is the audience? What is the service? What tone and length? Write your new prompt below (or on paper).

Version A — With laptop/internet:

Type your improved prompt into any AI chat tool at the table. Run it. Compare the output to what the weak prompt would have likely produced. Share your prompt and result with a neighbour.

Version B — Paper only:

Write your improved prompt on paper, labelling each part: R / C / T / F. Exchange papers with a neighbour. Each person reads the other's prompt aloud and guesses what kind of answer it would produce.

Facilitator answer key is provided in Appendix C.

1.6 Tool Directory for Prompting

Tool	Type	Best for
ChatGPT (OpenAI)	Freemium	General-purpose chat, writing, brainstorming, image understanding
Claude (Anthropic)	Freemium	Long documents, careful writing and analysis, coding help
Google Gemini	Freemium	Integrated with Gmail/Docs/Sheets, good for research with Google Search
Microsoft Copilot (Chat)	Freemium	Quick answers inside Windows and Edge, Microsoft 365 integration
Perplexity	Freemium	Research questions that need up-to-date web sources and citations
DeepSeek	Free	General chat and reasoning tasks, strong at a low cost
Mistral Le Chat	Freemium	Strong European alternative, good multilingual support
Meta AI (Llama)	Free	Built into WhatsApp, Instagram and Facebook

Qwen (Alibaba)	Free / Open	Strong open-weight model, good at coding and reasoning
Regional / sovereign LLMs (e.g. rinna, NTT "tsuzumi," SB Intuitions)	Enterprise	Local-language-tuned models for enterprise and government use

Key Takeaways — Session 1

- An AI model predicts likely next words based on patterns learned from huge amounts of text — it has no memory unless you give it context, and it can sound confident while being wrong.
- R-C-T-F (Role, Context, Task, Format) is a simple, reliable structure for any important prompt.
- Five techniques improve results: zero-shot, few-shot (show examples), chain-of-thought ("think step by step"), role prompting, and iterative refinement.
- Prompting is a loop, not a one-shot command — expect to refine two or three times.
- Always review AI output for facts, numbers and tone before using it for real work.

02

SESSION 2 · DAY 1, 10:45–12:15

AI for Office Productivity

Using AI inside the tools you already use every day

BY THE END OF THIS SESSION, YOU WILL BE ABLE TO

- Describe how an AI office assistant (Copilot, Gemini, Notion AI) fits into your everyday apps.
- Use AI prompts to draft, summarise, and polish documents, spreadsheets, slides, and email.
- Summarise a meeting and draft follow-up actions using AI.
- Identify at least three free or low-cost office productivity AI tools to try this month.

2 · AI for Office Productivity

2.1 The New Office Workday

Microsoft's global *Work Trend Index* research has repeatedly found that knowledge workers spend a large share of their day on email, meetings and searching for information — time that could otherwise go to higher-value thinking. AI office assistants are designed to attack exactly this problem: drafting the first version of a document, summarising something long, or turning a rough idea into a polished slide, in seconds rather than tens of minutes.

Adoption has moved quickly worldwide. According to Microsoft, roughly **85% of Nikkei 225 companies** (one of the world's major stock indices) now use Microsoft 365 Copilot in some form. **Sumitomo Corporation**, a global trading and investment group, became the first company in its sector to deploy Microsoft 365 Copilot company-wide, and has reported annual cost savings in the order of **\$8 million** from efficiency gains. **Dentsu**, the international advertising and media group, reports that Copilot automation saves employees up to **60 minutes per day**. These are not small pilot projects — they are large, public, verifiable deployments.

THE MINDSET SHIFT

Stop thinking "I need to write this." Start thinking "I need to review and improve a draft." AI's job is to remove the blank page, not to replace your judgement.

2.2 How an AI Office Assistant Works

Whether it is called Microsoft 365 Copilot, Google Gemini for Workspace, or Notion AI, the underlying pattern is the same.

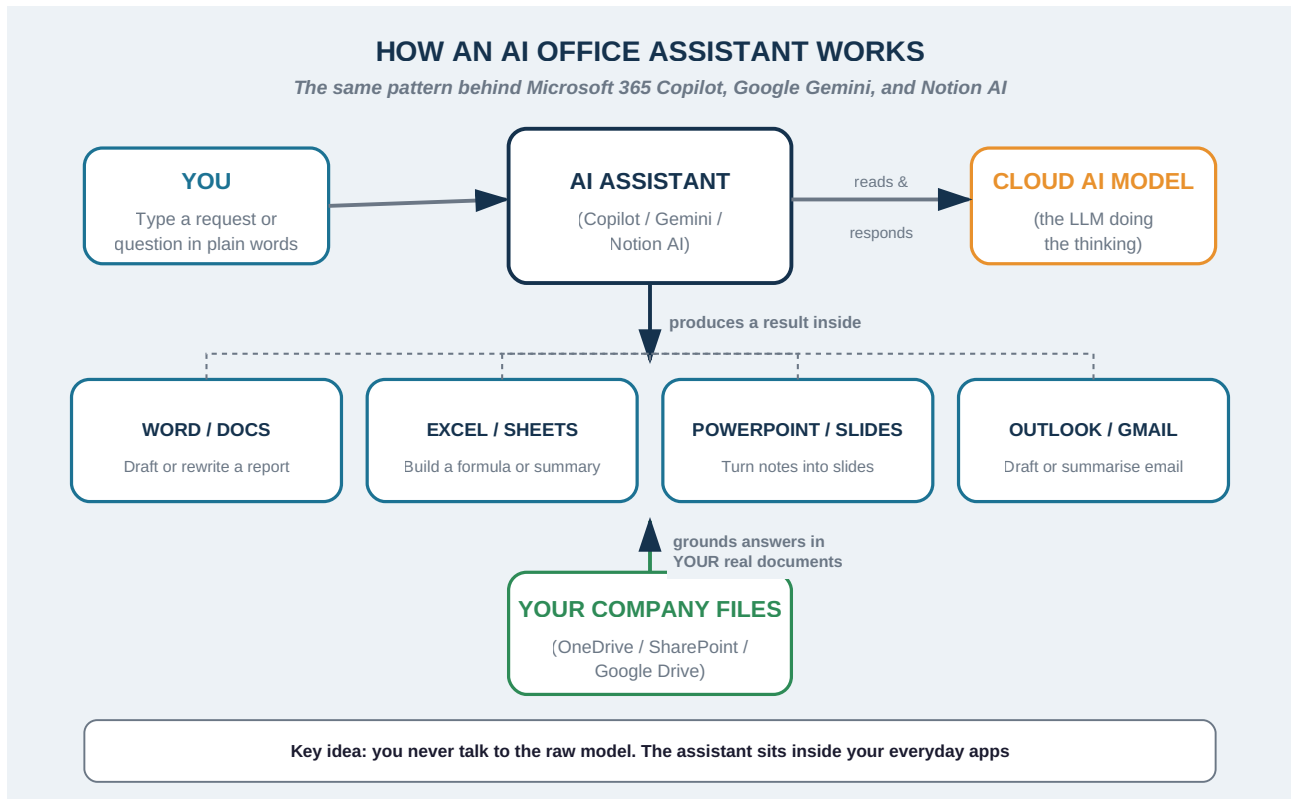


Figure 2.1 — How an AI office assistant connects you, your documents, and the underlying AI model.

Three things make this pattern powerful for office work, compared to a plain chat tool:

- **It is grounded in your own files.** Because it can read documents you have permission to access (with your organisation's security policy respected), it can answer "summarise last week's sales meeting notes" instead of only general knowledge questions.
- **It lives inside the app you are already using.** You do not copy text out to a separate chat window — you click a Copilot or AI icon directly inside Word, Excel, PowerPoint, Outlook, or Gmail.
- **It respects your organisation's access permissions.** It will not show you a file you do not already have permission to open. (Facilitators: this is worth repeating — it is the single most common security question from new users.)

2.3 AI Inside Word, Excel & PowerPoint

Word / Google Docs — drafting and rewriting

Use AI to produce a first draft, then edit it yourself. Good prompts specify audience and purpose:

TRY THIS PROMPT

"Rewrite this paragraph for a senior executive audience. Keep it under 80 words. Remove technical jargon. Keep a confident, factual tone." *(then paste your paragraph)*

Excel / Google Sheets — formulas and summaries

AI is excellent at writing formulas from a plain description, and at summarising a dataset's key patterns in words.

TRY THIS PROMPT

"I have a column of monthly sales figures in B2:B13. Write an Excel formula to calculate the percentage growth from the previous month for each row, and explain it in one sentence."

PowerPoint / Google Slides — turning notes into slides

Paste rough notes or bullet points and ask AI to structure them into a slide outline, then let the app's AI design feature (such as PowerPoint Designer or Copilot's "Create presentation from file") build the visual layout.

TRY THIS PROMPT

"Turn these five bullet points into a 6-slide outline: one title slide, one agenda slide, three content slides, one summary slide. Keep each slide to a maximum of 4 bullet points."

2.4 AI for Email and Meetings

Email — Outlook / Gmail

Common high-value uses: drafting a reply to a long email thread, shortening a long email before you send it, and adjusting tone (for example, making an email more polite and less direct, which is often useful when writing in English for a global audience).

Meetings — Teams / Zoom / Otter.ai

Meeting AI tools can transcribe a meeting live, then generate a summary with decisions and action items, even for participants who could not attend. One internal Microsoft example is instructive: inside Microsoft's own support team, Copilot reduced the time to summarise a customer enquiry from **5–10 minutes down to about 5 seconds**, reducing both workload and human error.

A PRACTICAL NOTE FOR MULTILINGUAL MEETINGS

Most meeting AI tools (Teams, Zoom AI Companion, Otter.ai) now support transcription and summarisation in many languages, and some can translate a summary from one language into another. This is particularly useful for meetings between teams based in different countries, where one summary can serve every audience.

Think about it

1. Which single task — email, meeting notes, a report, or a spreadsheet — would save you the most time this month if you used AI for it?

2.5 Real-World Case Studies

Amazon — AI for internal documents at scale

Amazon has described publicly how its own generative AI tools answer large numbers of internal employee and developer questions automatically, reducing time spent searching internal documentation and freeing staff for higher-value work — the same pattern available to any organisation through Microsoft 365 Copilot, Google Gemini, or Notion AI connected to internal files.

Sumitomo Corporation — a company-wide deployment

Sumitomo Corporation, a global trading and investment group, was among the first large organisations to roll out Microsoft 365 Copilot across its entire company, after an Early Access Program validation phase. Leadership has publicly attributed annual savings in the order of **\$8 million** to efficiency gains from the rollout — a concrete, board-level example of office AI ROI.

JCB — meeting people where they work

Global payments company JCB chose Microsoft 365 Copilot specifically because employees already used Word, Excel and PowerPoint daily — meaning the "AI icon" simply appeared inside tools people already trusted, rather than asking them to learn a brand-new system. JCB also built an internal "Generative AI Utilization Hub" to share good prompts between teams — a simple idea any organisation in this room could copy.

EXERCISE 2 Turn Notes Into a Polished Output

Duration: 20 minutes • Individual or pairs

The scenario: Below are rough meeting notes from a product planning meeting:

"Discussed Q2 launch. Marketing wants more budget. Engineering says timeline is tight, maybe slip 2 weeks. Sales is confident about demand in the North and South regions. Need decision from Director by Friday. Action: Priya to prepare budget options, Sam to check engineering timeline."

Version A — With laptop/internet:

Using R-C-T-F, write a prompt that turns these notes into (a) a 3-line executive summary and (b) a clear action-item table with owner and deadline. Run it in any AI tool. Compare results with a neighbour.

Version B — Paper only:

By hand, rewrite the notes into a 3-line executive summary and an action-item table. Then write the prompt you *would* have used to get AI to do this for you, labelling R / C / T / F.

2.6 Tool Directory for Office Productivity

Tool	Type	Best for
Microsoft 365 Copilot	Paid add-on	Word, Excel, PowerPoint, Outlook, Teams — deep integration
Google Gemini for Workspace	Paid add-on	Docs, Sheets, Slides, Gmail, Meet
Notion AI	Freemium	Notes, wikis, project docs with built-in AI search
Grammarly	Freemium	Grammar, tone and clarity checking for any written English
Otter.ai	Freemium	Live meeting transcription and automatic summaries
Zoom AI Companion	Included/Paid tiers	Meeting summaries and smart recaps inside Zoom
Canva (Magic Write / Magic Design)	Freemium	Quick slide and graphic design from a text prompt
DeepL	Freemium	High-quality translation between many language pairs, including documents
Slack AI	Paid add-on	Channel recaps and searchable answers inside Slack
Rakuten AI for Business	Enterprise	AI tools tailored for retail and e-commerce operations

Key Takeaways — Session 2

- AI office assistants sit inside the tools you already use (Word, Excel, PowerPoint, Outlook, Teams) — you rarely need a separate chat window.
- They are grounded in your own permitted files, which makes them more useful — and safer — than a general chat tool for company-specific questions.
- Real deployments (Sumitomo Corporation, JCB, Dentsu) show measurable, large-scale savings, not just small pilots.
- The highest-value starting points are usually: drafting/rewriting documents, building Excel formulas, turning notes into slides, and summarising meetings.
- Always review AI-drafted content for tone, accuracy and company policy before sending it externally.

03

SESSION 3 · DAY 2, 08:45–10:30

AI for Software Development & AI-Assisted Coding

How AI changes the way software gets built, tested and shipped

BY THE END OF THIS SESSION, YOU WILL BE ABLE TO

- Identify where AI assists each stage of the software development lifecycle.
- Explain the daily workflow of a developer using an AI pair-programming tool.
- Describe real productivity results from Amazon, eBay and Mercari.
- Recognise the main categories of AI coding tools, free and paid.

3 · AI for Software Development & AI-Assisted Coding

3.1 Why Coding Is Changing

You do not need to be a software engineer to understand why this topic matters for your organisation. Writing software has traditionally been slow and expensive, with a large share of time spent on repetitive, well-understood tasks: writing boilerplate code, writing tests, searching documentation, and fixing small bugs. AI coding assistants directly target these repetitive tasks, which is why adoption has moved so quickly inside engineering teams worldwide.

Three numbers illustrate the scale of the shift. The 2025 Stack Overflow Developer Survey found that **82% of developers who use AI tools use ChatGPT**, and **68% use GitHub Copilot** — meaning AI-assisted coding has gone from experimental to mainstream in only a few years. GitHub Copilot alone now has more than **26 million users worldwide** across virtually every country with an active software industry.

A NOTE FOR NON-ENGINEERS IN THE ROOM

Even if you do not write code yourself, understanding this topic helps you manage technical teams, evaluate vendor proposals, and have an informed conversation with engineering partners and clients.

3.2 AI Across the Development Lifecycle

A common misconception is that "AI coding" only means autocomplete while typing. In reality, AI now assists nearly every stage of building software.

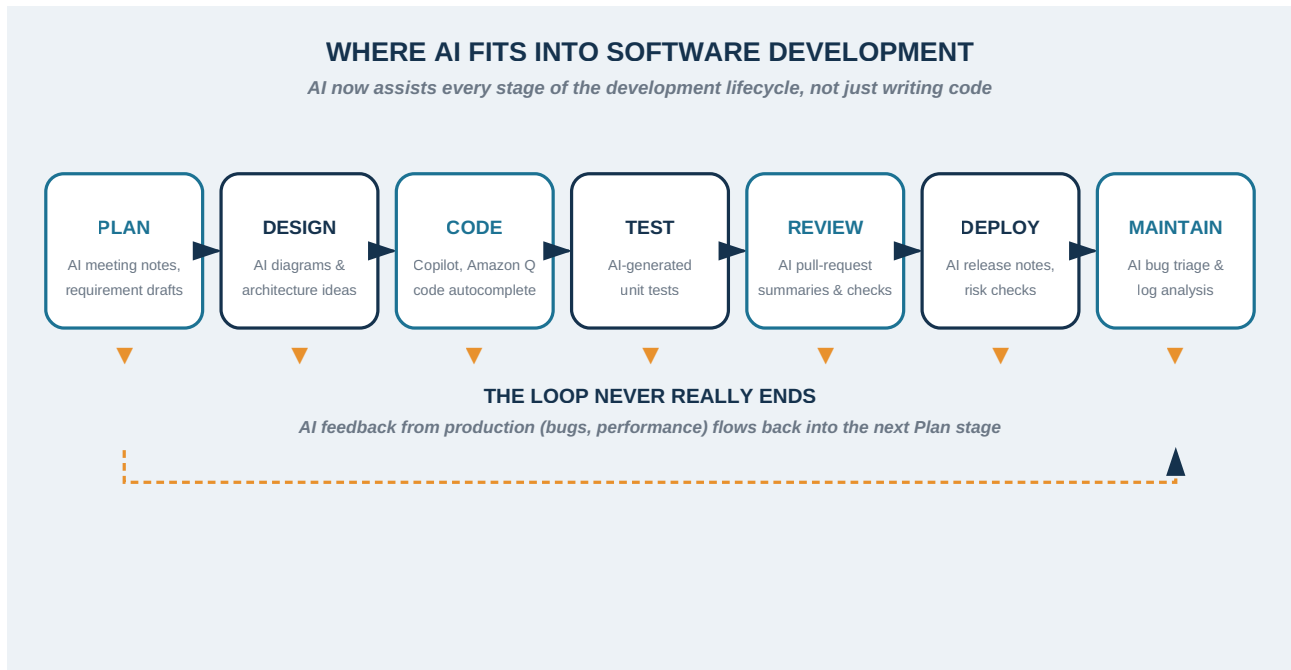


Figure 3.1 — Where AI fits into each stage of the software development lifecycle.

Plan & Design

AI drafts requirement documents from meeting notes, and can suggest a first-pass system architecture or diagram from a plain-language description.

Code

The most well-known use: AI suggests the next few lines, or a whole function, as a developer types, and can write a new function from a one-line natural-language instruction.

Test

AI can generate unit tests for existing code automatically, which is traditionally one of the most time-consuming and most frequently skipped parts of development.

Review, Deploy & Maintain

AI can summarise a pull request for a human reviewer, flag likely bugs or security issues before a human even looks at the code, draft release notes, and help triage production errors by reading log files and suggesting likely causes.

AMAZON Q DEVELOPER — A LARGE-SCALE EXAMPLE

Amazon disclosed that its internal AI coding agent helped migrate roughly **30,000 internal applications** from an old version of Java to a modern one — a task that would traditionally take a developer days per application — saving an estimated **4,500 developer-years of work and \$260 million per year**. In controlled trials, developers using Amazon Q Developer were **27% more likely** to successfully complete a coding task.

3.3 The AI-Assisted Coding Workflow

Here is what actually happens, step by step, when a developer uses a tool like GitHub Copilot or Amazon Q Developer during a normal working day.

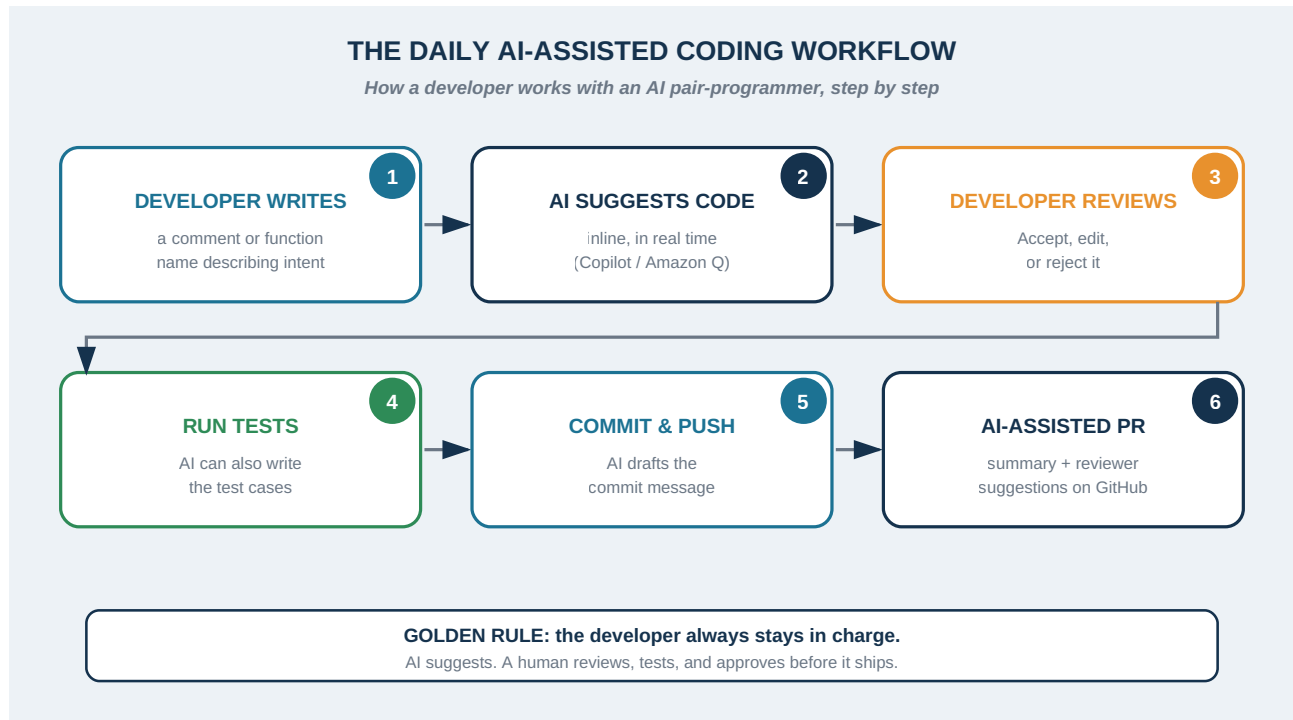


Figure 3.2 — The daily AI-assisted coding workflow, from intent to a reviewed pull request.

Notice that a human reviews the output at **every single stage**. This is the most important idea in this entire session: **AI assists, it does not replace, the developer's judgement**. The developer still decides what to build, reviews every suggestion, and is responsible for what gets shipped.

What a developer actually writes to "prompt" an AI coding tool

The same R-C-T-F thinking from Session 1 applies directly to coding. Instead of a paragraph, it is often a comment:

EXAMPLE: A ONE-LINE PROMPT AS A CODE COMMENT

```
// Function to validate a postal code (format: 12345 or 12345-6789)
and return true/false
```

The AI tool reads this comment as Context + Task, and suggests a working function immediately below it. The developer then reviews, tests, and edits as needed.

Common everyday use cases

- **Autocomplete** — finishing a line or a whole function as you type.
- **Explain this code** — useful when joining a new team or reading unfamiliar legacy code.
- **Find the bug** — pasting an error message and asking AI to suggest likely causes.
- **Write the tests** — generating unit tests for a function that currently has none.
- **Translate between languages** — converting a function from one programming language to another (for example, Python to JavaScript).

3.4 Real-World Case Studies

Amazon — Q Developer at massive scale

As introduced in Section 3.2, Amazon's internal use of its own Q Developer tool is one of the largest publicly disclosed AI-coding success stories in the industry: tens of thousands of applications modernised, with savings measured in thousands of developer-years and hundreds of millions of dollars annually.

eBay — engineering productivity alongside the Magical Listing tool

Beyond the customer-facing "Magical Listing" tool covered in Session 1, eBay's own engineering organisation has been a long-standing adopter of AI developer tools as part of its broader generative AI strategy, reflecting a pattern seen across most large e-commerce platforms: the same company that uses AI to help customers also uses AI to help its own engineers move faster.

Mercari — a consumer marketplace, company-wide AI coding

Mercari — a major consumer-to-consumer marketplace app, often compared to eBay — has rolled out both **GitHub Copilot and Cursor** to its entire engineering organisation of roughly 2,000 people, with both tools available for every engineer to use. Mercari's engineering leadership has spoken publicly about building an "AI-native" engineering culture, where every engineer understands what AI can and cannot do, and actively looks for new ways to apply it to daily work — not just as an isolated tool, but as a change in working habits.

WHY MERCARI IS A USEFUL REFERENCE POINT

Mercari faces the same scale challenges as eBay (millions of listings, fraud detection, search relevance) while operating as a consumer marketplace at national scale. Its public engineering talks are a good source of practical lessons on rolling out AI coding tools to a large team.

EXERCISE 3 Write the Comment, Predict the Code

Duration: 20 minutes • Pairs recommended

The scenario: You do not need to know how to code to do this exercise — you are practising how to describe a task clearly enough for an AI (or a junior colleague) to build correctly.

Your task: Write a one-sentence instruction (as if it were a code comment) for one of these tasks:

- Check whether a phone number is in a valid format.
- Calculate the total price of an online order, including an 8% shipping fee and 10% consumption tax.
- Sort a list of customer names alphabetically, ignoring upper/lower case.

Version A — With laptop/internet:

Type your instruction as a comment into any AI chat tool (e.g., "Write a Python function that... [your instruction]"). Read the generated code's comments/explanation (you do not need to understand every line) and check: does the explanation match what you asked for?

Version B — Paper only:

Swap your written instruction with a neighbour. Each of you writes, in plain English, the step-by-step logic you would expect a program to follow based on the other's instruction. Compare: did you both understand the same task?

3.5 Tool Directory for AI-Assisted Coding

Tool	Type	Best for
GitHub Copilot	Freemium	In-editor autocomplete + chat; covered in full in Session 4
Amazon Q Developer	Freemium	AWS-centric projects; strong at code migration/transformation
Cursor	Freemium	An AI-first code editor; deep whole-codebase understanding
Claude Code	Paid	Autonomous, CLI-based coding agent for larger, multi-step tasks
Codeium / Windsurf	Freemium	Free-friendly alternative autocomplete and chat
Tabnine	Freemium	Privacy-focused completion, can run on private infrastructure
Replit AI	Freemium	Learning to code, and building/deploying small apps in the browser

JetBrains AI

Freemium

Built into IntelliJ, PyCharm and other JetBrains IDEs

Key Takeaways — Session 3

- AI now assists nearly every stage of software development, not just typing code — planning, testing, review, deployment and maintenance all benefit.
- The daily workflow still places a human in charge at every step: write intent, AI suggests, human reviews, tests run, human commits.
- Real, large-scale results exist: Amazon saved thousands of developer-years and hundreds of millions of dollars using its own AI coding agent.
- Mercari, a major consumer-to-consumer marketplace, has deployed both GitHub Copilot and Cursor company-wide to around 2,000 engineers.
- You do not need to code to describe a task clearly — the same structured-instruction skill from Session 1 applies directly here.

04

SESSION 4 · DAY 2, 10:45–12:15

GitHub Copilot

A deep dive into the world's most widely used AI coding assistant

BY THE END OF THIS SESSION, YOU WILL BE ABLE TO

- Explain what GitHub Copilot is and how it technically works, end to end.
- Describe the Copilot product family: completions, Chat, Pull Requests, Workspace.
- Set up Copilot step by step in a code editor.
- Compare Copilot's plans and choose the right one for an individual or organisation.
- Describe real adoption results from NTT Docomo, Fujitsu and Sony.

4 • GitHub Copilot

4.1 What GitHub Copilot Is

GitHub Copilot, launched by GitHub (owned by Microsoft) and OpenAI in 2021, was the first mainstream AI coding assistant and remains the most widely used today, with more than **26 million users worldwide**. It works as an extension inside your code editor, suggesting code as you type and answering questions in a chat panel, trained on a model built for both natural language and programming languages.

Unlike a general chat tool, Copilot is purpose-built for the developer's daily environment: it reads your open files and project structure for context automatically, instead of requiring you to paste code into a separate window.

4.2 How Copilot Works Under the Hood

Understanding the architecture helps you reason about what Copilot can and cannot see, which is a common question from security and compliance teams.

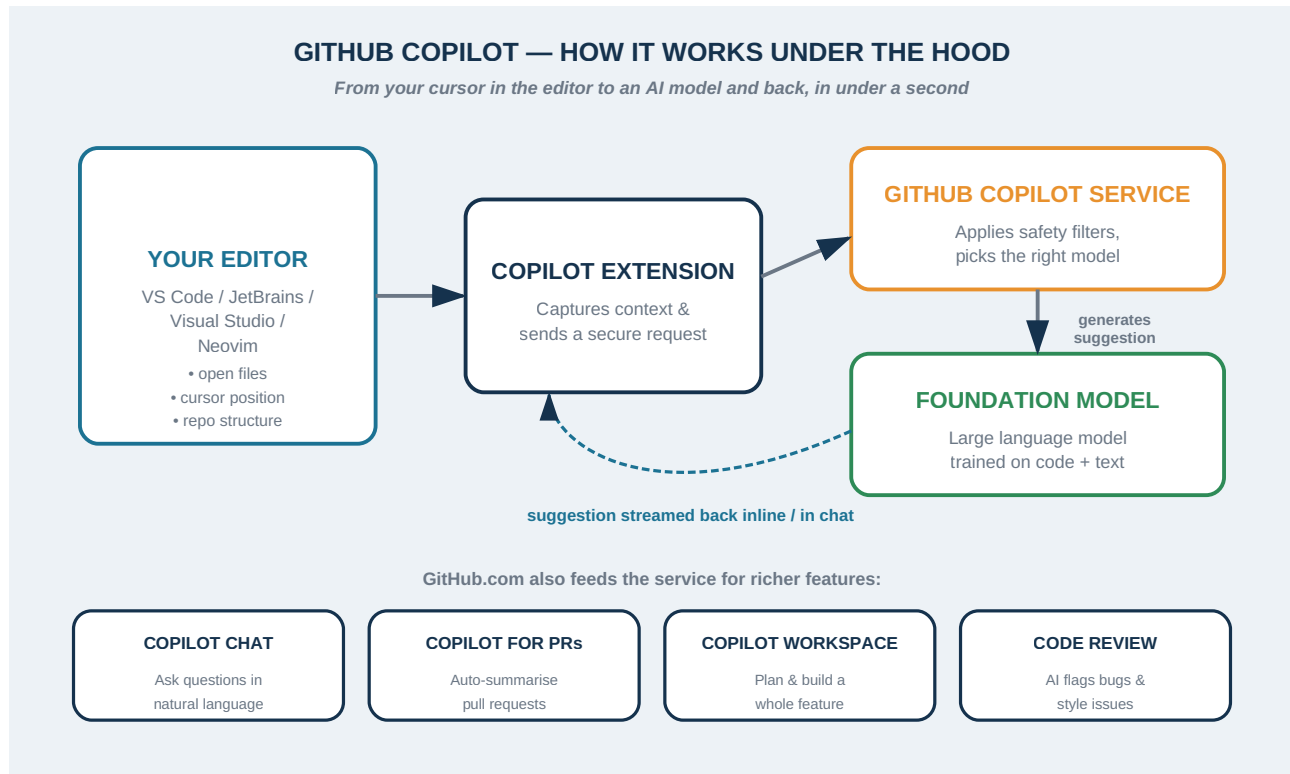


Figure 4.1 — GitHub Copilot's architecture, from your editor to the foundation model and back.

1. Your editor (VS Code, JetBrains IDEs, Visual Studio, or Neovim) captures context: open files, cursor position, and relevant parts of your repository.
2. The Copilot extension packages this context and sends a secure request to the GitHub Copilot service.
3. The service applies safety and content filters, and routes the request to the right underlying foundation model (Copilot today supports multiple model choices, including models from OpenAI, Anthropic and Google, depending on your plan).
4. The model generates a suggestion, which streams back through the service to your editor, appearing inline or in the chat panel, typically in well under a second.

A COMMON COMPLIANCE QUESTION, ANSWERED

For Business and Enterprise plans, GitHub contractually commits that your organisation's code is **not used to train** Copilot's underlying models. This is usually the single most important sentence to share with a security or legal team evaluating adoption.

4.3 The Copilot Product Family

"GitHub Copilot" is no longer a single feature — it is a family of related tools, all shown together in Figure 4.1:

- **Code Completion** — the original feature: inline suggestions as you type, from a single line to a whole function.
- **Copilot Chat** — a conversational panel inside your editor (and on github.com) for asking questions, explaining code, or requesting changes in natural language.
- **Copilot for Pull Requests** — automatically generates a plain-language summary of what changed and why, helping reviewers understand a change faster.
- **Copilot Workspace** — a higher-level, "agentic" feature that can plan and implement an entire small feature or bug fix across multiple files, with the developer approving the plan before changes are made.
- **Copilot Code Review** — an AI reviewer that flags likely bugs, security issues, or style problems automatically before a human reviewer looks at the change.

4.4 Getting Started, Step by Step

This is a practical walkthrough you, or anyone on your team, can follow today.

1. **Create or sign in to a GitHub account** at github.com (free).
2. **Choose a plan.** Start with the Free tier to try it — no credit card required — then upgrade to Pro, Business or Enterprise as needed (see Section 4.5).
3. **Install your editor's Copilot extension.** In VS Code: open the Extensions panel, search "GitHub Copilot," click Install. JetBrains IDEs, Visual Studio, and Neovim have equivalent extensions/plugins.
4. **Sign in to Copilot** from inside the editor — this opens a browser window to securely link your GitHub account.
5. **Start typing.** Suggestions appear automatically, shown in grey "ghost text." Press **Tab** to accept, or keep typing to ignore.
6. **Open Copilot Chat** (usually a sidebar icon or keyboard shortcut) to ask questions in plain, natural language, in whichever language you are most comfortable with.
7. **Try the built-in slash commands** inside chat: `/explain` (explain selected code), `/fix` (propose a fix for an error), `/tests` (generate unit tests), `/doc` (write documentation for a function).

YOUR FIRST FIVE MINUTES WITH COPILOT CHAT

Open any existing file in your project and type into Copilot Chat: *"Explain what this file does, in simple terms, for someone new to the project."* This single prompt is one of the fastest ways to get comfortable navigating an unfamiliar codebase.

4.5 Plans and Pricing

GitHub Copilot's pricing moved to a usage-based "AI Credits" model in mid-2026, but the familiar tier structure remains the easiest way to compare options.

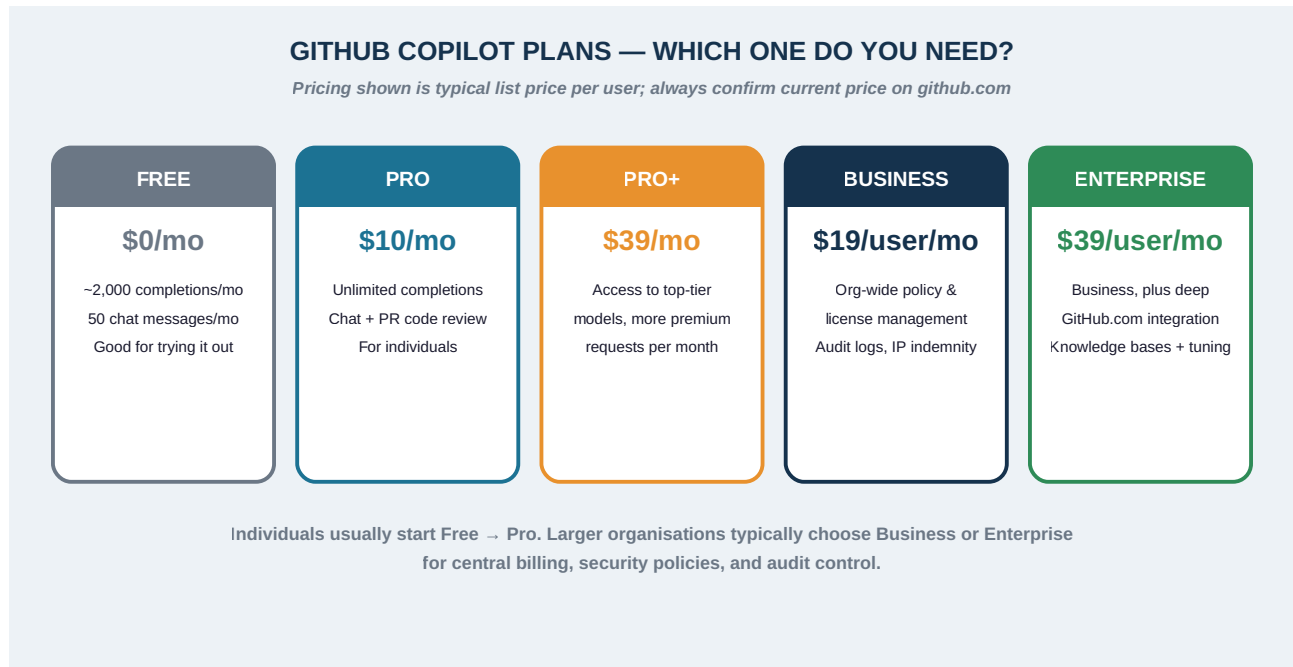


Figure 4.2 — GitHub Copilot's plans, from Free to Enterprise. Confirm current prices at github.com/features/copilot/plans before budgeting.

Individuals typically start on Free (around 2,000 completions and 50 chat messages per month, at no cost) and move to Pro (\$10/month) once they rely on it daily. Power users running Copilot as an always-on coding agent for hours a day may choose Pro+ or the Max tier.

Organisations almost always choose Business (\$19/user/month) or Enterprise (\$39/user/month, requires GitHub Enterprise Cloud) for centralised billing, usage policies, audit logs, and the data-protection guarantee described in Section 4.2.

4.6 Real-World Case Studies

NTT Docomo — measurable daily usage at scale

NTT Docomo's development teams have used GitHub Copilot since March 2023. As of a November 2025 GitHub Japan briefing, **3,647 developers** use it, with roughly **1,000 people actively using it on an average day**. Docomo reported receiving more than **30,000 code suggestions per day**, of which about **25% are accepted** into real production work — a concrete, transparent adoption metric that any engineering manager can relate to.

Fujitsu — a measured pilot with a clear result

Fujitsu ran a pilot of GitHub Copilot alongside Microsoft Dev Box with **2,400 developers**. The company's own technology leadership reported that developers using the combination **reduced their workload by 20%**, while maintaining Fujitsu's strict security requirements for protecting customer data.

Sony and ZOZO — further confirmed adopters

GitHub has publicly named **Sony** and fashion e-commerce company **ZOZO** among confirmed enterprise adopters of GitHub Copilot, reflecting adoption across industries — electronics, telecom, and retail — not just "pure" technology companies.

A PATTERN WORTH NOTICING

Every case study in this book shares one feature: adoption was measured carefully (active users, acceptance rate, workload reduction) rather than adopted on faith — good practice for any organisation piloting Copilot.

EXERCISE 4 Plan a Copilot Rollout

Duration: 20 minutes • Small groups of 3–4

The scenario: Your organisation is considering GitHub Copilot for a 50-person engineering team.

Version A — With laptop/internet:

Visit github.com/features/copilot and, as a group, decide: (1) which plan fits best and why, (2) one metric you would track in month one (borrow an idea from NTT Docomo's approach), (3) one security question you would ask before rollout.

Version B — Paper only:

As a group, write your answers to the same three questions using only the pricing table in Figure 4.2 and the case studies in Section 4.6. Present your group's rollout plan to the room in two minutes.

Key Takeaways — Session 4

- GitHub Copilot connects your editor, a context-aware service layer, and a foundation model — suggestions return in under a second.
- It is a family of tools: Completions, Chat, Pull Requests, Workspace, and Code Review, not just one feature.
- Getting started takes about five minutes: install the extension, sign in, and start typing.
- Business and Enterprise plans guarantee your code is not used to train Copilot's models — a key fact for compliance conversations.
- Real-world deployments (NTT Docomo, Fujitsu, Sony, ZOZO) show measurable, significant results at enterprise scale.



APPENDIX A

Master AI Tool Directory

A larger reference list, for work and for everyday life — free and paid options side by side

Appendix A • Master AI Tool Directory

This appendix gathers every tool mentioned in this book, plus additional everyday-life tools, into one reference. Prices and features change frequently — always confirm on the vendor's website before purchasing or recommending a paid plan to your organisation.

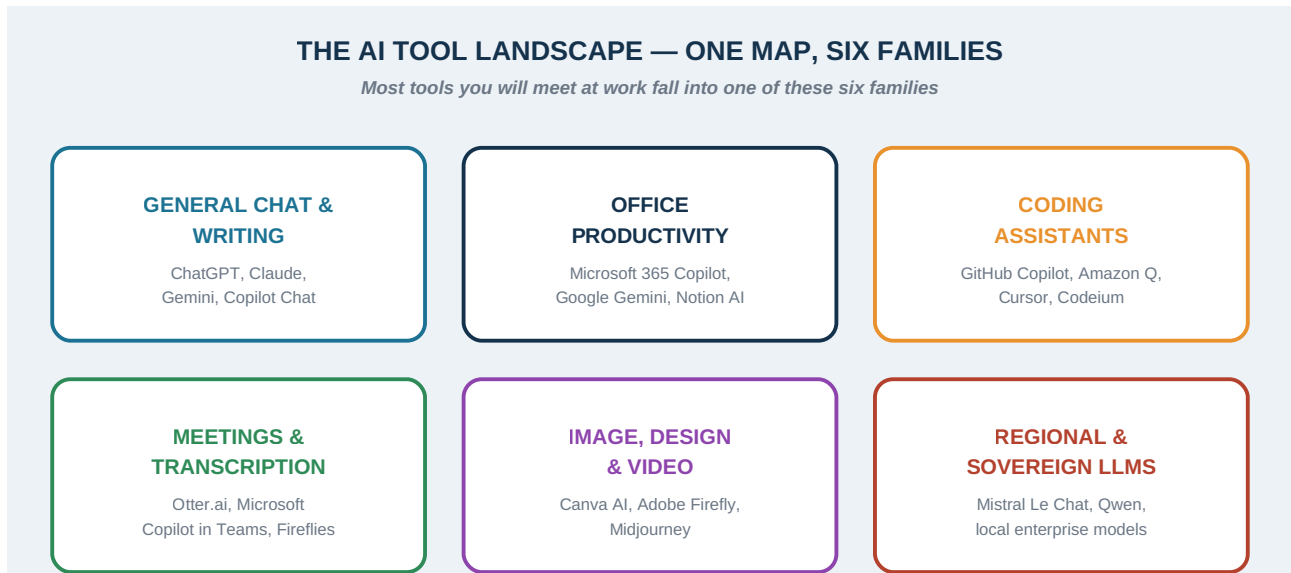


Figure A.1 — The AI tool landscape, organised into six practical families.

A.1 General Chat, Writing & Research

Tool	Type	Notes
ChatGPT	Freemium	Most widely used general AI assistant; strong all-rounder
Claude	Freemium	Careful, detailed writing and analysis; strong with long documents
Google Gemini	Freemium	Deep Google Search and Workspace integration
Microsoft Copilot (Chat)	Freemium	Built into Windows and Edge browser
Perplexity	Freemium	Research assistant with cited, up-to-date web sources
DeepSeek	Free	Strong reasoning at very low/no cost
DeepL	Freemium	Best-in-class translation across many language pairs
Grammarly	Freemium	English grammar, tone and clarity checking

A.2 Office Productivity & Collaboration

Tool	Type	Notes
Microsoft 365 Copilot	Paid add-on	Word, Excel, PowerPoint, Outlook, Teams
Google Gemini for Workspace	Paid add-on	Docs, Sheets, Slides, Gmail, Meet
Notion AI	Freemium	Notes, wikis, and project knowledge bases
Otter.ai	Freemium	Meeting transcription and summaries
Zoom AI Companion	Included/ Paid	Meeting recaps and smart summaries
Fireflies.ai	Freemium	Alternative meeting recorder/summariser
Canva (Magic Write/Design)	Freemium	Slides, graphics and social media content
Slack AI	Paid add-on	Channel recaps and searchable answers inside Slack
Rakuten AI for Business	Enterprise	AI tools tailored to retail and e-commerce operations

A.3 Coding & Software Development

Tool	Type	Notes
GitHub Copilot	Freemium	Covered in full detail in Session 4
Amazon Q Developer	Freemium	Strong for AWS projects and legacy code migration
Cursor	Freemium	AI-first code editor; whole-codebase understanding
Claude Code	Paid	Autonomous coding agent for larger, multi-step tasks
Codeium / Windsurf	Freemium	Free-friendly completion and chat
Tabnine	Freemium	Can run on private infrastructure for data sensitivity
Replit AI	Freemium	Learn to code and ship small apps in the browser
JetBrains AI	Freemium	Built into IntelliJ, PyCharm and other JetBrains IDEs

A.4 Image, Design & Video

Tool	Type	Notes
Canva AI	Freemium	Posters, slides, social graphics from a text prompt
Adobe Firefly	Freemium	Image generation integrated into Photoshop/Illustrator
Midjourney	Paid	High-quality artistic image generation
Synthesia	Paid	AI avatar video generation for training/marketing videos
CapCut	Freemium	AI-assisted video editing, popular for short-form video

A.5 Regional & Local-Language Models

Tool	Type	Notes
Mistral Le Chat	Freemium	Leading European model, strong multilingual support
Qwen (Alibaba)	Free/Open	Strong open-weight model, widely used across Asia
Meta AI (Llama)	Free	Built into WhatsApp, Instagram and Facebook
Sovereign/enterprise LLMs (e.g. NTT "tsuzumi," SB Intuitions, rinna)	Enterprise	Local-language-tuned models for enterprise/government use

A.6 Everyday Life — Beyond the Office

Tool	Type	Notes
Google Translate	Free	Instant translation and camera-based text translation while travelling
Duolingo Max	Freemium	AI-powered language learning with conversational practice
Google/Apple Maps AI directions	Free	AI-assisted trip and route planning
Perplexity (personal use)	Freemium	Everyday research: health, travel, product comparisons
	Freemium	

ChatGPT / Gemini (personal use)		Meal planning, trip itineraries, personal letters, learning new topics
Photomath	Freemium	AI help with children's or personal maths homework

HOW TO CHOOSE, QUICKLY

Start free. Most people only need one paid subscription (often a general chat tool like ChatGPT Plus or Claude Pro) plus whatever their employer provides (Microsoft 365 Copilot or GitHub Copilot). Add a specialist tool only when a specific, recurring task justifies its cost.

Appendix B · Glossary of AI Terms

AI Credits

GitHub Copilot's usage-based billing unit, introduced in June 2026, used alongside or instead of flat monthly limits.

Autocomplete (code completion)

AI suggesting the next piece of code as a developer types, shown as grey "ghost text" until accepted.

Chain-of-thought prompting

Asking an AI model to reason step by step before giving a final answer, which reduces careless errors on multi-step tasks.

Context window

The maximum amount of text (prompt, documents and reply combined) an AI model can consider at one time.

Few-shot prompting

Giving an AI model one or more examples of the desired output style before asking it to complete a new, similar task.

Foundation model

A large AI model (such as GPT, Claude, or Gemini) trained on vast amounts of text and code, which powers many different products.

Generative AI

AI that creates new content (text, images, code, audio) rather than only classifying or predicting from existing data.

GitHub Copilot Workspace

An "agentic" Copilot feature that plans and implements a feature or fix across multiple files, with developer approval at each step.

Hallucination

When an AI model generates incorrect or fabricated information in a fluent, confident-sounding way.

LLM (Large Language Model)

An AI model trained to predict and generate human language (and often code) based on patterns learned from large text datasets.

Prompt

The instruction or question a person gives to an AI model.

Prompt engineering

The practice of writing clear, structured prompts to reliably get useful, accurate AI output.

Pull request (PR)

A proposed set of code changes submitted for review before being merged into a shared project.

R-C-T-F framework

A simple prompt structure covering Role, Context, Task and Format — used throughout this book.

Token

A small unit of text (roughly a word or part of a word) that an AI model processes; usage and pricing are often measured in tokens.

Zero-shot prompting

Asking an AI model to complete a task directly, with no examples provided.

C

APPENDIX C

Facilitator Notes & Answer Keys

Minute-by-minute pacing guides and model answers for every exercise

Appendix C · Facilitator Notes & Answer Keys

C.1 Minute-by-Minute Pacing

Session 1 — Prompt Engineering (105 minutes)

Minutes	Segment
0–10	Welcome, icebreaker, why prompt engineering matters (1.1)
10–25	How a language model works (1.2)
25–45	The R-C-T-F framework, with Figure 1.1 (1.3)
45–65	Five core techniques, with Figure 1.2 (1.4)
65–80	Case studies: Amazon, eBay, Rakuten (1.5)
80–95	Exercise 1: Fix the Weak Prompt
95–105	Tool directory, key takeaways, Q&A

Session 2 — AI for Office Productivity (90 minutes)

Minutes	Segment
0–8	Why this matters, adoption statistics (2.1)
8–20	How an AI office assistant works, Figure 2.1 (2.2)
20–40	AI in Word, Excel, PowerPoint with live or guided examples (2.3)
40–55	AI for email and meetings (2.4)
55–65	Case studies: Amazon, Sumitomo, JCB (2.5)
65–85	Exercise 2: Turn Notes Into a Polished Output
85–90	Tool directory handout, key takeaways

Session 3 — AI for Software Development (105 minutes)

Minutes	Segment
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0–10	Why coding is changing, adoption statistics (3.1)
10–30	AI across the development lifecycle, Figure 3.1 (3.2)
30–50	The AI-assisted coding workflow, Figure 3.2 (3.3)
50–65	Case studies: Amazon, eBay, Mercari (3.4)
65–85	Exercise 3: Write the Comment, Predict the Code
85–100	Tool directory and open discussion
100–105	Key takeaways, bridge to Session 4

Session 4 — GitHub Copilot (90 minutes)

Minutes	Segment
0–8	What GitHub Copilot is (4.1)
8–25	Architecture and product family, Figure 4.1 (4.2, 4.3)
25–40	Live or guided "Getting Started" walkthrough (4.4)
40–55	Plans and pricing, Figure 4.2 (4.5)
55–70	Case studies: NTT Docomo, Fujitsu, Sony (4.6)
70–85	Exercise 4: Plan a Copilot Rollout, with group presentations
85–90	Whole-course wrap-up and final Q&A

C.2 Model Answers — Exercise 1 (Fix the Weak Prompt)

A strong rewrite might read: "You are a marketing writer for a technology company. Our corporate clients are evaluating cloud security services and value clear, trustworthy, low-hype communication. Write a 150-word announcement introducing our new managed security service, explaining what problem it solves and one concrete benefit. Use a formal, respectful tone suitable for an international business audience, and structure it as a short paragraph followed by three bullet points."

Discussion point: ask participants which word in the original prompt ("something") was the single biggest problem, and why.

C.3 Model Answers — Exercise 2 (Turn Notes Into a Polished Output)

Executive summary: "The Q2 launch is on track but faces two open risks: a possible 2-week engineering delay and a marketing budget request. Sales reports strong confidence in Osaka and Nagoya demand. A Director decision is needed by Friday."

Action table: Tanaka-san – prepare budget options – before Friday decision. Sato-san – confirm engineering timeline – before Friday decision.

C.4 Model Answers — Exercise 3 (Write the Comment, Predict the Code)

Example instruction: "Write a function that checks if a phone number is valid. It should accept formats like 555-123-4567 or 5551234567, and return true or false." A good AI response will explain, in plain language, that it checks the number of digits and the overall pattern — participants do not need to read the code itself, only confirm the explanation matches their intent.

C.5 Model Answers — Exercise 4 (Plan a Copilot Rollout)

Plan fit: For a 50-person team with centralised billing and audit needs, Business (\$19/user/month) is the usual starting point; upgrade to Enterprise only if deep GitHub.com integration or custom fine-tuning is required.

Metric to track: weekly active users and suggestion-acceptance rate, following NTT Docomo's public example.

Security question: "Can we confirm, in writing, that our proprietary code will not be used to train Copilot's models?" (Answer: yes, contractually, for Business and Enterprise plans.)

C.6 General Facilitation Tips

- Run every exercise's Version A (with laptops) if connectivity allows — seeing a real AI response live is more memorable than any slide.
- If the room has mixed English confidence, pair stronger and less confident English speakers together for exercises.
- Keep a visible timer during exercises; professional audiences generally respond well to clear, respected time-boxes.
- End each session by asking one participant to summarise, in their own words, the single most useful idea — this reinforces retention far better than a facilitator-led recap alone.

END OF COURSE MATERIAL

Thank you for learning with us

We hope this book continues to be useful long after the classroom sessions end. Keep it as a reference as AI tools continue to evolve.